

ALFIDO TECH AI INTERNSHIP

Task 1 — ML Classification Project

SMS Spam Detection Using Machine Learning and NLP

Prepared as part of the Alfido Tech Artificial Intelligence Internship

1. Project Overview

The objective is to classify SMS messages as Ham (legitimate) or Spam using supervised machine learning. The workflow includes cleaning, exploratory analysis, TF-IDF feature extraction, stratified splitting, cross-validation, model comparison, evaluation, and model serialization.

2. Dataset

The SMS Spam Collection dataset contains 5,572 labeled messages. The labels were encoded as Ham = 0 and Spam = 1. Duplicate rows were removed.

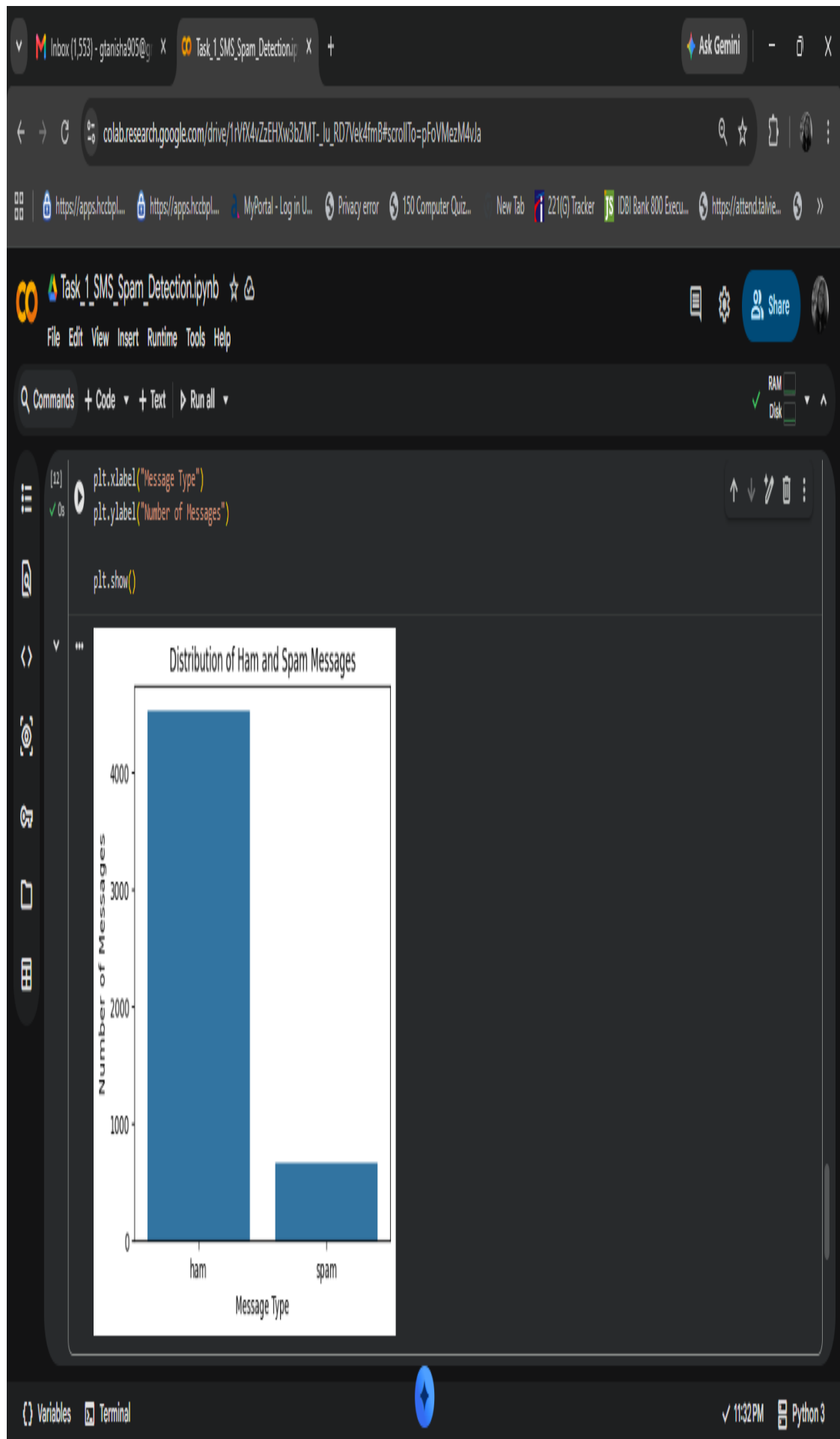


Figure 1. Distribution of Ham and Spam messages.

3. Exploratory Data Analysis

The class distribution is imbalanced, with substantially more Ham messages than Spam messages. Message length was also analyzed.

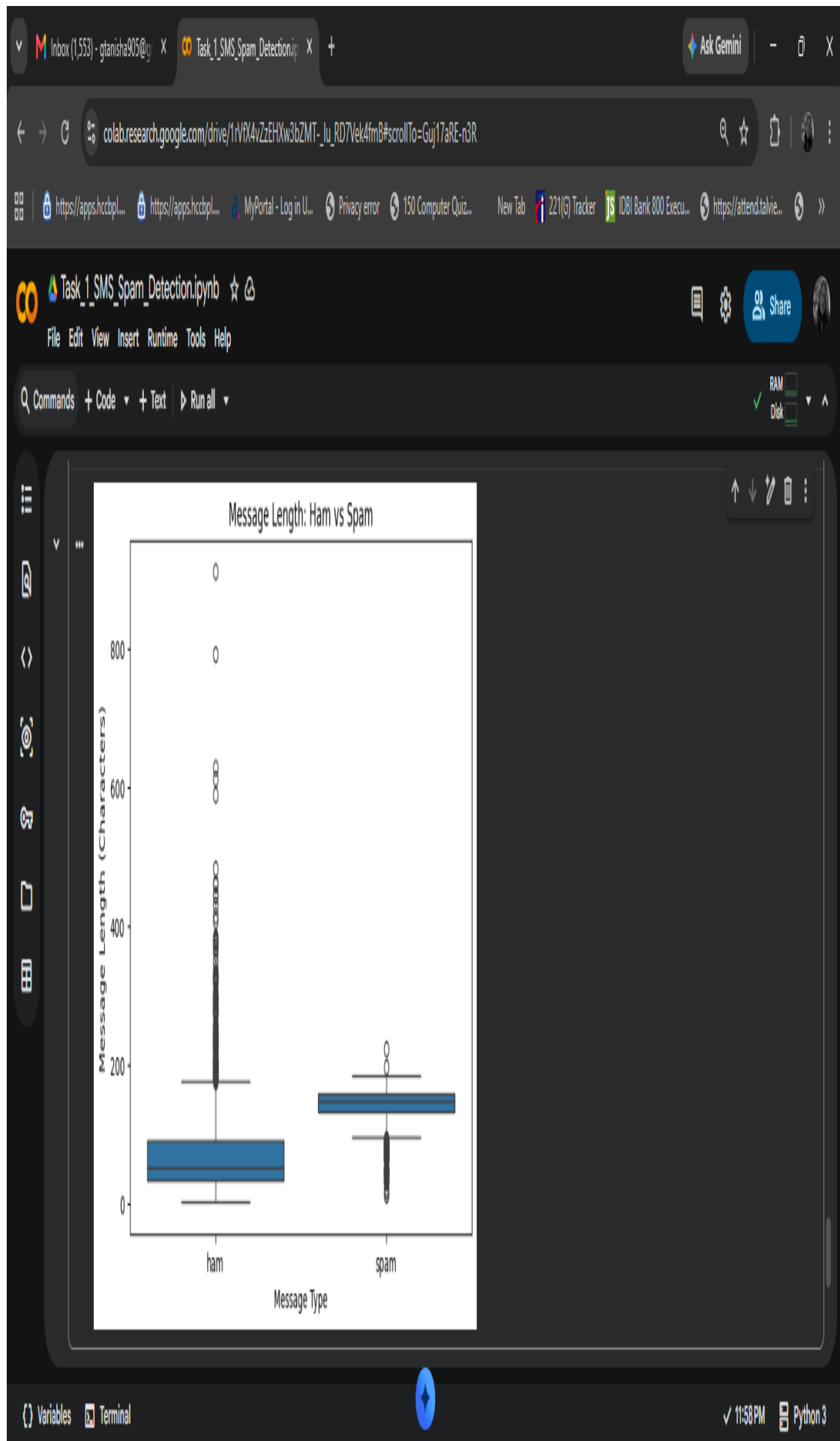


Figure 2. Message length comparison.

4. Preprocessing and Feature Engineering

An 80:20 stratified train/test split was used. TF-IDF with lowercase text, English stop-word removal, unigrams and bigrams, and a maximum vocabulary of 5,000 features converted SMS text into numerical features. TF-IDF was fitted only on training data for the initial test evaluation and was placed inside a pipeline for cross-validation.

5. Algorithms

Logistic Regression was used as a baseline classifier. Random Forest was used as the second algorithm with balanced class weights.

6. Cross-Validation

Metric	Logistic Regression Mean	Logistic Regression Std	Random Forest Mean	Random Forest Std
Accuracy	0.9543	0.0069	0.9717	0.0057
Precision	0.9850	0.0093	0.9926	0.0100
Recall	0.6476	0.0496	0.7816	0.0411
F1-Score	0.7806	0.0375	0.8741	0.0274
ROC-AUC	0.9875	0.0030	0.9867	0.0038

Random Forest showed stronger mean cross-validation performance on accuracy, precision, recall and F1-score. Logistic Regression had a marginally higher mean ROC-AUC.

Task_1_SMS_Spam_Detection.ipynb

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rt_cv_results["test_roc_auc"].std()
    ]
})

display(cv_summary.round(4))
```

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	Metric	Logistic Regression Mean	Logistic Regression Std	Random Forest Mean	Random Forest Std
0	Accuracy	0.9543	0.0069	0.9717	0.0057
1	Precision	0.9850	0.0093	0.9926	0.0100
2	Recall	0.6476	0.0496	0.7816	0.0411
3	F1-Score	0.7806	0.0375	0.8741	0.0274
4	ROC-AUC	0.9875	0.0030	0.9867	0.0038

Variables

Terminal

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7. Test-Set Results

Metric	Logistic Regression	Random Forest
Accuracy	0.9565	0.9797
Precision	0.9778	1.0000
Recall	0.6718	0.8397
F1-Score	0.7964	0.9129
ROC-AUC	0.9920	0.9936

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rf_f1,
rf_roc_auc

]

})

print("FINAL MODEL COMPARISON")
print("=" * 50)

display(final_results.round(4))
```

FINAL MODEL COMPARISON

	Metric	Logistic Regression	Random Forest
0	Accuracy	0.9665	0.9797
1	Precision	0.9778	1.0000
2	Recall	0.6718	0.8397
3	F1-Score	0.7964	0.9129
4	ROC-AUC	0.9920	0.9936

Variables Terminal

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8. Model Selection

Random Forest was selected as the final model. On the held-out test set it achieved 97.97% accuracy, 100.00% precision, 83.97% recall, 91.29% F1-score and 99.36% ROC-AUC. Its overall balance of required metrics was stronger than Logistic Regression.

9. Custom Prediction

A promotional message was classified as Spam with 71.67% spam probability, while a normal meeting message was classified as Ham with 0.33% spam probability.

10. Conclusion

The project demonstrates a complete supervised NLP classification workflow and produces a reusable Random Forest model and TF-IDF vectorizer for later deployment. The next stage can expose the saved model through an API and containerize it.